Tutorial Sheet-06

1. Let X and Y be dependent random variables with common means 0, variances 1, and correlation coefficient ρ. Show that
$$E[\max(X^2, Y^2)] < 1 + \sqrt{1-\rho^2}.$$

2. Let (X, Y) be jointly distributed with probability density function (PDF) f defined by:
$$f(x,y) = \begin{cases} \frac{1}{2}, & \text{inside the square with corners at } (0,1), (1,0), (-1,0), (0,-1), \\ 0, & \text{otherwise.} \end{cases}$$
(Note: The region is defined by $|x| + |y| \leq 1$).
Are X and Y independent? Are they uncorrelated?

3. Let X_1 and X_2 be independent discrete random variables following symmetric discrete uniform distributions:
$$X_1 \sim \text{Uniform}\{-n, -n+1, \ldots, n\} \quad \text{and} \quad X_2 \sim \text{Uniform}\{-m, -m+1, \ldots, m\},$$
where n and m are positive integers.
Let two new random variables Z and W be defined by the transformation:
$$Z = X_1 \cos\theta + X_2 \sin\theta \quad \text{and} \quad W = X_2 \cos\theta - X_1 \sin\theta.$$
Find the correlation coefficient, $\rho_{Z,W}$, between Z and W, and show that:
$$|\rho_{Z,W}| \leq \frac{|m(m+1) - n(n+1)|}{m(m+1) + n(n+1)}.$$

4. Suppose r balls are distributed independently into n cells such that each arrangement is equally likely.
   (a) Let X be the number of balls in a specific cell. Show that X follows a Binomial distribution and find the probability p_k that the cell contains exactly k balls.
   (b) Suppose that the number of balls r and the number of cells n are both very large, such that the average density of balls per cell remains constant (i.e., $r, n \to \infty$ with $\frac{r}{n} = \lambda$). Show that the probability found in part (a) converges to a Poisson probability mass function with parameter λ:
$$\lim_{n\to\infty} p_k = \frac{e^{-\lambda}\lambda^k}{k!}.$$

5. Show that the terms of the Poisson probability mass function (PMF) given by
$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \quad k = 0, 1, 2, \ldots,$$
reach their maximum when k is the largest integer less than or equal to λ (denoted as $k = \lfloor\lambda\rfloor$). Furthermore, show that if λ is an integer, the maximum is reached at two points: $k = \lambda - 1$ and $k = \lambda$.

6. Let $X_1, X_2, \ldots, X_n$ be independent Poisson random variables such that $X_i \sim P(\lambda_i)$ for $i = 1, \ldots, n$. Let $S_n = \sum_{i=1}^n X_i$ be their sum, and let $\Lambda = \sum_{i=1}^n \lambda_i$.
   Derive the conditional joint probability mass function of the random vector $(X_1, X_2, \ldots, X_n)$ given that the total sum is fixed at $S_n = m$.
   $$P(X_1 = k_1, X_2 = k_2, \ldots, X_n = k_n \mid S_n = m)$$
   where $k_i \geq 0$ are integers such that $\sum_{i=1}^n k_i = m$.

7. Vehicles arrive at a toll plaza from two independent feeder roads, Road A and Road B. The arrival of vehicles from Road A follows a Poisson process with a rate of $\lambda_A = 2$ vehicles/minute, while vehicles from Road B arrive with a rate of $\lambda_B = 3$ vehicles/minute.
   (a) Suppose it is observed that exactly $n = 10$ vehicles arrived at the toll plaza in a specific one-minute interval. Let X be the number of vehicles that came from Road A.
   Using the conditional property of independent Poisson variables, determine the probability distribution of X. Specify its parameters.
   (b) Calculate the probability that exactly 4 of these 10 vehicles came from Road A.
   (c) The toll charges are different for the two roads: vehicles from Road A pay Rs. 50, while vehicles from Road B pay Rs. 80 (due to heavier trucks). Given that 10 vehicles arrived in total, calculate the expected total revenue collected during this one minute interval.

8. The arrival of customer support calls at a tech support center is modelled as a Poisson process with a rate of $\lambda = 3$ calls per minute.
   (a) The support team considers a 2-minute interval to be manageable if they receive exactly 5 calls. Calculate the probability of this event.
   (b) A senior engineer needs 30 seconds of uninterrupted time to restart a server. What is the probability that *zero* new calls arrive during this specific 30-second interval?
   (c) Is it more likely to receive exactly 1 call in a 20-second interval or exactly 2 calls in a 40-second interval?